1. EC2 (Elastic Compute Cloud)

- **Purpose**: Scalable virtual servers in the cloud for running applications.
- **Key Points**:
  - Customizable configurations (CPU, memory, storage, networking).
  - Flexible pricing models: On-demand, Reserved, and Spot Instances.
  - Auto Scaling for dynamic resource adjustment based on traffic.
  - Supports Linux, Windows, and custom AMIs.
  - Integrates with VPC, S3, and IAM for better security and networking.

---

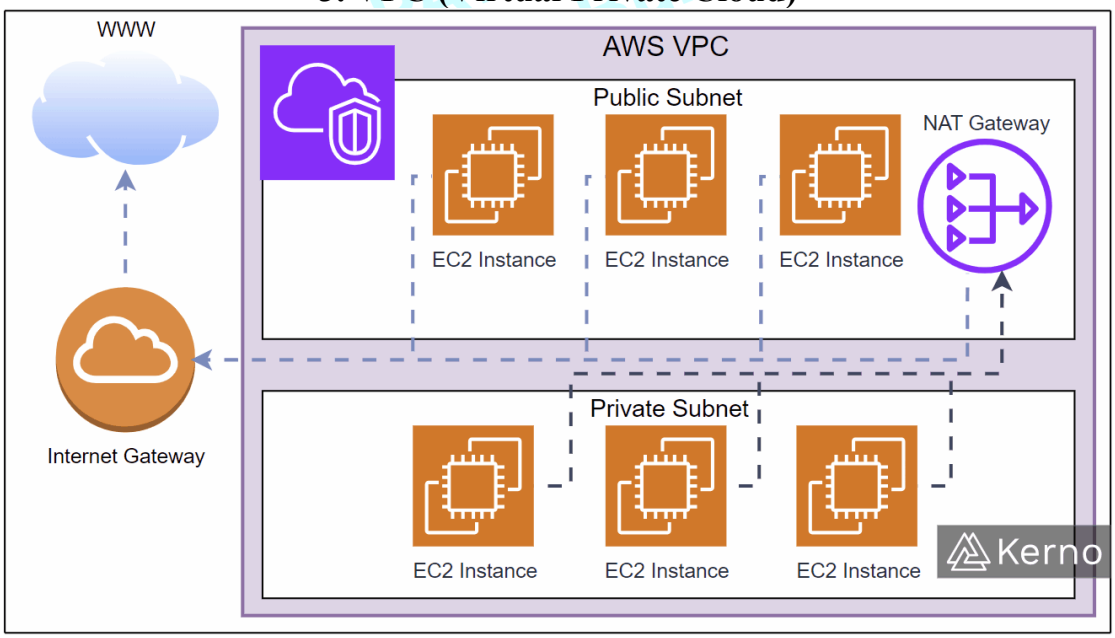
2. IAM (Identity and Access Management)

- **Purpose**: Secure management of access to AWS resources.
- **Key Points**:
  - Creates users, groups, roles, and policies to control permissions.
  - Supports MFA (Multi-Factor Authentication) for enhanced security.
  - Fine-grained access control for different AWS resources.
  - Centrally manages and audits user activities.
  - Works with AWS services to manage permissions across all resources.

PATHNEX
Shaping Careers, Building Futures

---

3. VPC (Virtual Private Cloud)



- **Purpose**: Creates isolated networks in AWS for resources.
- **Key Points**:
  - Customizable IP ranges, subnets, and route tables.
  - Provides secure resource communication with security groups and NACLs.
  - Supports VPN and Direct Connect for hybrid cloud setups.
  - Multi-AZ deployment for high availability.

- Ensures fault tolerance and security for resources.

---

4. S3 (Simple Storage Service)

- **Purpose**: Scalable object storage for storing and retrieving data.
- **Key Points**:
  - Virtually unlimited, durable, and cost-effective storage.
  - Multiple storage classes for varying access needs.
  - Built-in encryption for data at rest and in transit.
  - Versioning for object management and restoration.
  - Easily integrates with Lambda, CloudFront, and other AWS services.

---

5. Session Manager

- **Purpose**: Secure shell access to EC2 instances without needing SSH/RDP.
- **Key Points**:
  - Remote management without the need for opening inbound ports.
  - Logging and auditing for session activities.
  - Allows access to EC2 instances in private subnets.
  - Integrates with AWS Systems Manager for automation.
  - No need for public IP addresses on instances.

PATHNEX
Shaping Careers, Building Futures

---

6. Route 53

- **Purpose**: DNS service for routing traffic to resources.
- **Key Points**:
  - Domain registration and DNS routing management.
  - Configures routing policies like weighted, latency-based, and geolocation.
  - Health checks and DNS failover to improve availability.
  - Integrates with CloudFront and ELB for dynamic routing.
  - Offers fast DNS resolution for applications.

---

7. CloudTrail

- **Purpose**: Monitors and records AWS API calls for auditing.
- **Key Points**:
  - Tracks changes made to AWS resources and user activities.
  - Logs are stored in S3 for security and analysis.
  - Helps with compliance and auditing for regulatory requirements.
  - Integrates with CloudWatch for real-time monitoring.

- Provides detailed logs for troubleshooting security incidents.

---

8. CloudWatch

- **Purpose**: Monitoring and observability service for AWS resources.
- **Key Points**:
  - Collects logs, metrics, and events from AWS resources.
  - Set up alarms to trigger automated actions or notifications.
  - Helps visualize resource performance and system health.
  - Allows for centralized log storage and analysis.
  - Integrates with Auto Scaling and Lambda for automated management.

---

9. EKS (Elastic Kubernetes Service)

- **Purpose**: Managed Kubernetes service for containerized applications.
- **Key Points**:
  - Automates the deployment and scaling of containerized applications.
  - Provides integration with AWS services like IAM, VPC, and CloudWatch.
  - Supports multi-cluster and multi-region architectures.
  - Built-in security features like IAM roles for service accounts.
  - Integrates easily with DevOps workflows and CI/CD pipelines.

PATHNEX
Shaping Careers, Building Futures

---

10. Auto Scaling

- **Purpose**: Automatically adjusts resources to meet demand.
- **Key Points**:
  - Scales EC2 instances, and other resources based on real-time needs.
  - Ensures cost-efficiency and high availability of applications.
  - Works with Elastic Load Balancer (ELB) to balance traffic.
  - Supports scaling across multiple availability zones.
  - Provides automated scaling for ECS and Lambda functions.

---

11. Lambda

- **Purpose**: Serverless compute service that runs code in response to events.
- **Key Points**:
  - No need to provision or manage servers.
  - Supports multiple languages like Node.js, Python, Java, and Go.
  - Pay only for the compute time your code uses.
  - Can be triggered by events from S3, DynamoDB, SNS, API Gateway, and more.

- Ideal for event-driven architectures and microservices.

---

12. AWS Shield

- **Purpose**: DDoS protection for AWS services.
- **Key Points**:
  - Shield Standard automatically protects AWS customers.
  - Shield Advanced offers enhanced protection, including real-time attack visibility.
  - Protects resources like CloudFront, Route 53, and ELB.
  - Provides 24/7 DDoS response team (DRT) support.
  - Helps mitigate large-scale attacks without affecting performance.

---

13. AWS WAF (Web Application Firewall)

- **Purpose**: Protects web applications from common security threats.
- **Key Points**:
  - Customizable rules to filter malicious web traffic.
  - Protects against SQL injection, cross-site scripting, and bots.
  - Integrates with CloudFront, ALB, and API Gateway.
  - Provides real-time logging of web traffic patterns.
  - Automates security responses using AWS Lambda.

PATHNEX Shaping Careers, Building Futures

---

14. Amazon GuardDuty

- **Purpose**: Threat detection service for AWS environments.
- **Key Points**:
  - Continuously monitors AWS accounts for suspicious activity.
  - Uses machine learning and threat intelligence to detect anomalies.
  - Provides actionable findings to address security risks.
  - Integrates with CloudWatch for automatic response actions.
  - Helps identify unauthorized access, reconnaissance, and data exfiltration.

---

15. Amazon Inspector

• **Purpose:** Automated vulnerability management service for AWS workloads.

- **Key Points:**

  - Continuously scans AWS workloads (EC2 instances, container images, and Lambda functions) for security vulnerabilities.

- Automatically discovers resources and assesses them against known CVEs and security best practices.
- Provides detailed findings with severity ratings and remediation guidance.
- Integrates with Amazon EventBridge and CloudWatch for automated alerts and response workflows.
- Helps identify software vulnerabilities, unintended network exposure, and deviations from security best practices.

---

16. AWS Key Management Service (KMS)

- **Purpose**: Manages encryption keys for securing data within AWS.
- **Key Points**:
  - Supports symmetric and asymmetric encryption keys.
  - Easily integrates with AWS services like S3, RDS, and EBS.
  - Automates key rotation to simplify key management.
  - Provides fine-grained access control for encryption keys.
  - Audits key usage via CloudTrail for compliance.

---

17. AWS Systems Manager (SSM)

PATHNEX
Shaping Careers, Building Futures

- **Purpose**: Automates operational tasks and manages AWS infrastructure.
- **Key Points**:
  - Manages patching, configurations, and resource compliance.
  - Supports remote management of EC2 instances and on-premises resources.
  - Automates routine administrative tasks like software deployments.
  - Provides centralized management with CloudWatch and CloudTrail.
  - Helps with security and compliance auditing.

---

18. Amazon RDS (Relational Database Service)

- **Purpose**: Managed relational database service for multiple database engines.
- **Key Points**:
  - Supports MySQL, PostgreSQL, Oracle, SQL Server, and MariaDB.
  - Automates backups, patch management, and failover.
  - Provides scalability through read replicas and multi-AZ deployments.
  - Offers encryption at rest and in transit for secure data management.
  - Integrates with CloudWatch for monitoring and performance metrics.

---

19. Amazon Aurora

- **Purpose**: High-performance relational database compatible with MySQL and PostgreSQL.
- **Key Points**:
  - Provides up to 5x the performance of MySQL and 2x that of PostgreSQL.
  - Self-healing storage and automated backups.
  - Aurora Serverless scales automatically based on demand.
  - Multi-AZ deployments for high availability.
  - Fully managed with automatic patching and scaling.

---

20. Amazon DynamoDB

- **Purpose**: Managed NoSQL database service for high-performance applications.
- **Key Points**:
  - Provides fast, low-latency read and write operations.
  - Automatically scales throughput and storage.
  - Supports both key-value and document data models.
  - Built-in encryption at rest for security.
  - Integrates with Lambda, CloudWatch, and other AWS services.

---

21. Amazon Redshift

- **Purpose**: Managed data warehouse service for running complex queries.
- **Key Points**:
  - Provides fast SQL-based querying for structured data.
  - Columnar storage and parallel query processing.
  - Scales with Redshift Spectrum for querying data in S3.
  - Integrates with AWS analytics services like Glue and EMR.
  - Offers encryption for data security.

---

22. AWS DMS (Database Migration Service)

- **Purpose**: Helps migrate databases to AWS with minimal downtime.
- **Key Points**:
  - Supports migrations between homogeneous and heterogeneous databases.
  - Continuous data replication to ensure data integrity.
  - Monitors migration progress and provides detailed reports.
  - Enables migration of on-premises databases to AWS.
  - Supports many database engines, including Oracle, MySQL, and SQL Server.

PATHNEX
Shaping Careers, Building Futures

23. AWS Batch

- **Purpose**: Efficiently runs batch processing workloads at scale.
- **Key Points**:
  - Automatically provisions the compute resources for batch jobs.
  - Handles thousands of batch computing jobs at any scale.
  - Supports custom compute environments and job scheduling.
  - Integrates with S3 for data storage and processing.
  - Supports EC2 Spot Instances for cost optimization.

---

24. AWS Glue

- **Purpose**: Managed ETL service for data preparation and loading.
- **Key Points**:
  - Automatically crawls and discovers datasets.
  - Transforms, cleans, and prepares data for analytics.
  - Integrates with Amazon Redshift, S3, and other data sources.
  - Supports both Python and Scala for scripting.
  - Serverless architecture removes the need for infrastructure management.

---

25. Amazon Elastic File System (EFS)

- **Purpose**: Scalable file storage service for EC2 instances and on-premises resources.
- **Key Points**:
  - Provides scalable, shared file storage accessible by EC2 instances.
  - Fully managed with automatic scaling.
  - Supports NFS protocols for integration with on-premises environments.
  - Provides encryption at rest and in transit.
  - Ideal for use cases like content management and big data analytics.

---

26. AWS Elastic Beanstalk

- **Purpose**: Platform-as-a-Service (PaaS) for deploying and managing applications.
- **Key Points**:
  - Supports multiple programming languages, including Java, Python, Node.js, and more.
  - Automatically handles deployment, scaling, and monitoring.
  - Simplifies application management with an easy-to-use console.
  - Integrates with other AWS services like RDS, S3, and SNS.
  - Allows easy customization with configuration options.

PATHNEX
Shaping Careers, Building Futures

27. Amazon CloudFront

- **Purpose**: Content delivery network (CDN) for faster content delivery.
- **Key Points**:
  - Distributes content to users globally with low latency.
  - Supports dynamic and static content delivery.
  - Can serve from edge locations close to users for faster speeds.
  - Integrates with S3, EC2, and Lambda for dynamic content delivery.
  - Provides DDoS protection and encryption for security.

---

28. AWS Snowball

- **Purpose**: Data transport solution for securely transferring large amounts of data to AWS.
- **Key Points**:
  - Provides a physical appliance to transport data to AWS.
  - Useful for large-scale data migrations when internet transfer is impractical.
  - Integrates with S3 and other storage services for data import.
  - Supports encryption for secure data handling.
  - Offers Snowball Edge for local compute and storage capabilities.

PATHNEX
Shaping Careers, Building Futures

29. AWS CodePipeline

- **Purpose**: Continuous integration and delivery service for automated software workflows.
- **Key Points**:
  - Automates the build, test, and deployment process of applications.
  - Integrates with GitHub, CodeCommit, and other source control repositories.
  - Supports multiple stages for flexible pipeline configurations.
  - Can integrate with Lambda for serverless application deployments.
  - Provides full visibility into your CI/CD process.

---

30. AWS CodeBuild

- **Purpose**: Managed build service to compile source code and run tests.
- **Key Points**:
  - Compiles code from source control like GitHub and CodeCommit.
  - Runs tests on your code and produces build artifacts.
  - Scales automatically based on build volume.
  - Integrates with AWS CodePipeline and other CI/CD tools.
  - Supports multiple build environments like Java, Node.js, Python, and Go.

---

31. AWS CodeDeploy

- **Purpose**: Automates code deployment to EC2 instances and on-premises servers.
- **Key Points**:
  - Supports rolling, blue/green, and canary deployments.
  - Integrates with AWS services like EC2, Lambda, and ECS.
  - Provides detailed deployment logs for visibility.
  - Enables zero-downtime application deployments.
  - Supports integration with third-party CI/CD tools.

---

32. AWS X-Ray

- **Purpose**: Tracing service for debugging and analyzing microservices applications.
- **Key Points**:
  - Provides insights into the performance of distributed applications.
  - Helps identify performance bottlenecks and troubleshoot errors.
  - Supports integration with Lambda, EC2, and other services.
  - Visualizes application architecture and request paths.
  - Helps with security and compliance monitoring.

PATHNEX
Shaping Careers, Building Futures